

TRAINING WORKBOOK

Window Tint Installation

Core Course — 3-Day Beginner Program

Student Workbook · Version 1.0

Student Name: _____

Class Date: _____

Instructor: _____

Location: _____

Window Tint Installation — Student Workbook

Skill Forge Core Course · 3-Day Beginner Program

Welcome

Welcome to the Skill Forge Window Tint Core Course. Over three intensive days you will build the foundational knowledge and hands-on skills needed to install automotive window film professionally — from your first flat-glass cut to a complete vehicle tint job.

This workbook is your companion through classroom sessions, shop demonstrations, and supervised installs. Complete the knowledge checks, take notes during labs, and use the checklists to track your progress toward course completion.

What "Core" means: This course is designed for beginners with little or no tint experience. By Day 3 you will have installed film on multiple glass types and completed a supervised full-vehicle tint. Advanced techniques (ceramic specialty installs, commercial fleet work, windshield full-tint where permitted) are covered in our Advanced courses.

Pre-Course Checklist

Before You Arrive

- ☐ Review your state's window tint laws (VLT limits for front side, rear side, rear window, windshield)
- ☐ Wear closed-toe shoes and shop-appropriate clothing all three days
- ☐ Bring safety glasses if you have a preferred pair (loaners available)
- ☐ Bring a notebook and pen for additional notes beyond this workbook
- ☐ Remove rings and bracelets that can scratch glass during installs
- ☐ Trim fingernails — long nails make film handling difficult

Provided Each Day

- Practice glass panels and vehicle access for supervised installs
 - Window film (practice and project rolls)
 - Tools: squeegees, heat gun, spray bottles, blades, tint slip solution
 - Lunch and refreshments
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Course Schedule Overview

Day	Theme	Focus
Day 1	Foundations	Film science, laws, tools, safety, flat glass
Day 2	Technique	Shrinking, curved glass, door glass, troubleshooting
Day 3	Professional Practice	Rear window, full vehicle, QC, business basics, assessment

Day 1 — Foundations

Day 1 Learning Objectives

By the end of Day 1, you will be able to:

1. Identify common window film types and their performance characteristics
 2. Explain VLT (Visible Light Transmission) and how tint laws apply
 3. Set up a tint workstation with proper lighting and contamination control
 4. Perform a complete glass prep sequence on automotive glass
 5. Cut, position, and squeegee film on flat practice glass with minimal defects
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Day 1 Schedule

Time	Session
8:00 – 8:30	Welcome, introductions, course overview
8:30 – 9:30	Classroom: Window film types & performance
9:30 – 10:00	Classroom: Tint laws & customer consultation basics
10:00 – 10:15	Break
10:15 – 11:15	Classroom: Tools, chemicals, workspace setup
11:15 – 12:00	Classroom: Safety — heat, blades, chemicals, electrical (defrosters)
12:00 – 1:00	Lunch
1:00 – 1:30	Demo: Glass prep and contamination inspection
1:30 – 3:00	Lab 1: Flat glass cut, wet-out, and squeegee technique
3:00 – 3:15	Break
3:15 – 4:30	Lab 2: Pattern cutting from roll — inside vs outside cut
4:30 – 5:00	Day 1 knowledge check & Q&A

Session 1: Window Film Types & Performance

How Window Film Works

Automotive window film is a multi-layer laminate applied to the interior surface of glass. Key layers include:

- **Scratch-resistant hard coat** — protects the film surface from abrasion

- **Tinted or coated layer** — provides UV rejection, heat rejection, and visible light reduction
- **Adhesive layer** — bonds film to glass (pressure-sensitive or water-activated depending on product)
- **Release liner** — protects adhesive until installation

Common Film Categories

Type	Construction	Heat Rejection	Notes
Dyed	Dyed layer in polyester	Moderate	Economical; can fade over time in harsh sun
Metalized	Metal particles in film	Good	Can interfere with radio/GPS/cell signals
Hybrid	Dyed + metalized	Good	Balance of performance and cost
Carbon	Carbon particles	Very good	No signal interference; good color stability
Ceramic	Ceramic nano-particles	Excellent	Premium tier; best heat rejection, no signal issues

Installer tip: Learn to explain film tiers in customer language: "This film blocks 99% of UV rays" is more meaningful to most customers than quoting IR rejection percentages.

Key Performance Metrics

- **VLT (Visible Light Transmission)** — percentage of visible light passing through the film + glass assembly. Lower VLT = darker appearance.
- **UV rejection** — quality automotive films block 99%+ of UV rays regardless of VLT.
- **TSER (Total Solar Energy Rejected)** — overall heat rejection; important for upselling ceramic.
- **IRR (Infrared Rejection)** — measures rejection of infrared heat; ceramic films excel here.

Session 2: Tint Laws & Customer Consultation

Understanding VLT Limits

Every state (and many municipalities) regulates how dark window tint can be on each window position. Limits are expressed as minimum VLT — the percentage of light that must pass through.

Typical window positions regulated:

- Front driver & passenger (side)
- Front windshield (often only brow/visor strip allowed)
- Rear side windows
- Rear window (back glass)

Critical: Film VLT ratings are measured on the film alone. Installed VLT on glass is affected by the glass itself. Always consult current state law and your film manufacturer's installed VLT data before advising customers.

Customer Consultation Basics

Before any install, confirm:

1. **Legal limits** for their state and which windows they want tinted
2. **Film tier** — dyed, carbon, ceramic — and explain the performance difference
3. **Warranty** — film warranty vs. your shop's workmanship warranty
4. **Cure time** — windows stay up 24-72 hours depending on film and weather
5. **Existing tint** — old film must be removed before new install (additional charge)

Document everything. A signed work order with film type, VLT, windows to be tinted, and price prevents disputes.

Session 3: Tools, Chemicals & Workspace

Essential Tint Tools

Tool	Purpose
Film cutting knife	Trim film edges, pattern cutting
Hard card / squeegee set	Wet-out and remove slip solution
Soft squeegee (felt or low-friction)	Final passes on film surface
Heat gun	Shrink film on curved glass
Spray bottles	Slip solution and rinse
Scrub pad / steel wool (#0000)	Glass prep (product-specific)
Tint-safe blade holder	Clean edges on glass (use with care)
Heat gun holster / stand	Keep heat gun accessible but safe
Film plotter (shop)	Computer-cut patterns — introduced Day 3

Slip Solution Recipe (General)

Most shops use a mild soap-and-water mix:

- 1 quart distilled water
- 2-4 drops mild soap or your film supplier's recommended slip agent

Too much soap = film slides excessively and won't grab. **Too little** = film sticks before positioned.

Workspace Requirements

- **Clean floor** — lint and dust are the enemy
 - **Good lighting** — LED shop lights at the glass surface reveal contamination instantly
 - **Climate control** — ideal install temp 50-90°F; avoid dusty or windy conditions with doors open
 - **Organized film storage** — rolls upright, climate-controlled, away from dust
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Session 4: Safety

Non-negotiable safety rules for every install:

1. Heat guns reach 1,000°F+ — never point at skin, wiring, or interior trim
2. Razor blades cut glass AND skin — always cut away from your body
3. Ammonia-based cleaners can damage film adhesive and interior surfaces
4. Defroster lines are fragile — excessive blade pressure or heat can break them
5. Wear eye protection when using heat guns near shrinking film

Chemical Safety

- Know your SDS (Safety Data Sheet) for all cleaning products
 - Ventilate the workspace when using strong degreasers
 - Never mix chemicals unless the manufacturer specifies it
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Lab 1: Glass Prep & Flat Glass Install

Lab 1 — Flat Glass Prep & Install

Objective: Prepare a flat practice glass panel and install a piece of window film with clean edges and minimal contamination.

Time: 90 minutes

Part A — Glass Prep (20 min)

1. Inspect glass under shop lighting for existing contamination, adhesive residue, or defects
2. Spray with slip solution and scrub with tint-safe pad
3. Squeegee dry completely — water trapped under film causes bubbles
4. Final pass with lint-free cloth; inspect for remaining particles
5. **Checkpoint:** Glass should be squeaky-clean. Run your fingernail along the surface — if you feel grit, re-prepare.

Part B — Cut & Position (25 min)

1. Unroll film on the **outside** of the glass (liner facing you)
2. Cut film roughly 1" larger than glass on all sides
3. Peel liner partially; spray adhesive side with slip solution
4. Spray glass interior with slip solution
5. Position film on interior glass; align edges with slight overhang
6. **Checkpoint:** Film should slide freely for positioning

Part C — Wet-Out & Squeegee (35 min)

1. Starting from the center, squeegee outward in firm, overlapping strokes
2. Push solution and air to the edges — do not trap air in the center
3. Tack down edges; trim excess with fresh blade using glass as guide
4. Final light pass with soft squeegee
5. **Checkpoint:** No visible bubbles larger than a pinhead; edges fully adhered

Part D — Self-Assessment

Rate your install (1-5): Edge cleanliness ____ | Bubble count ____ | Contamination ____

Lab 2: Pattern Cutting from Roll

Lab 2 — Roll Pattern Cutting

Objective: Cut a pattern for a door glass using the outside-cut method.

Time: 60 minutes

Outside Cut Method

1. Place film on exterior glass with liner out
2. Trim excess to manageable size
3. Cut to glass shape using the glass edge as your guide
4. Peel liner, wet adhesive, transfer to interior
5. Wet-out and finish

Inside Cut Method (preview)

1. Place film inside glass, liner out
2. Cut to fit from interior — more precise but tighter workspace
3. Used commonly on rear windows and final trims

Practice both methods on your door glass assignment during Day 2.

Day 1 Knowledge Check

1. What does VLT stand for, and how does a lower VLT number affect appearance?
2. Name three differences between dyed film and ceramic film.
3. Why is distilled water preferred for slip solution?
4. What are three signs that glass is not clean enough for film application?
5. Why do we cut patterns on the outside of the glass but install on the inside?
6. What safety precautions apply when using a heat gun?

Day 1 Notes

Day 2 — Technique Development

Day 2 Learning Objectives

By the end of Day 2, you will be able to:

1. Explain the heat-shrink process and when it is required
 2. Shrink film on a curved practice panel using a heat gun
 3. Remove and reinstall a door glass (or work through door access method)
 4. Install film on a curved quarter window with clean edges
 5. Identify and correct common defects: fingers, bubbles, contamination, creases
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Day 2 Schedule

Time	Session
8:00 – 8:30	Day 1 review & Q&A
8:30 – 9:45	Classroom: Heat shrinking theory & technique
9:45 – 10:30	Demo: Shrinking on curved glass
10:30 – 10:45	Break
10:45 – 11:45	Classroom: Door glass — removal, access, defroster lines
11:45 – 12:00	Classroom: Troubleshooting common defects
12:00 – 1:00	Lunch
1:00 – 2:30	Lab 3: Curved panel shrink & install
2:30 – 2:45	Break
2:45 – 4:30	Lab 4: Door glass tint install (supervised)
4:30 – 5:00	Day 2 knowledge check & Q&A

Session 5: Heat Shrinking Theory

Why Shrink?

Flat film does not naturally conform to compound curves (bowed rear windows, curved door glass). **Shrinking** uses heat to relax the polyester film so it contracts and conforms to the curve without fingers (lifted areas that won't adhere).

The Shrink Process

1. Place film on the **outside** of the curved glass (liner out)
2. Identify low points and fingers that need shrinkage
3. Apply heat in sweeping motions — never hold stationary
4. Film will wrinkle, then tighten — stop before overheating
5. Smooth with gloved hand or squeegee; repeat until film lays flat
6. Cut pattern, then transfer to interior for install

Overheating causes: film discoloration, adhesive failure, and visible distortion. If you see rainbow haziness, you've gone too far. Start over with fresh film.

Heat Gun Technique

- Keep nozzle 4–6 inches from film surface
- Use continuous sweeping motion
- Shrink from the center of fingers toward edges
- Let film cool before cutting — shrinking continues slightly as it cools

Session 6: Door Glass & Defroster Lines

Door Glass Access Methods

Method	When to Use	Notes
Drop glass	Most modern vehicles	Lower glass 2–3" for bottom edge access
Panel removal	Tight seals, some trucks/SUVs	Remove door panel; protect clips and wiring
Full glass removal	Deep tint, problem installs	Requires urethane/seal knowledge — advanced

Working Around Defroster Lines

Rear windows and some side glass have defroster lines (conductive ceramic frit lines):

- **Never scrape across lines** with a blade
- Squeegee **parallel** to lines, not across them
- Use a soft squeegee with light pressure over lines
- Heat shrinking over lines requires extra care — lines are raised

Session 7: Troubleshooting Common Defects

Defect	Likely Cause	Fix
Finger	Insufficient shrink on curve	Remove film; re-shrink or start fresh
Bubble	Trapped air/solution	Small: pin and squeegee; large: remove and redo
Contamination	Dirty glass or dusty environment	Remove film, re-prep glass, reinstall
Crease	Film folded during handling	Cannot fix — replace film
Dust speck	Airborne particle during install	Pin if tiny; redo if visible
Edge lift	Incomplete squeegee at edge	Re-squeegee; heat lightly if needed

The 24-hour rule: Minor haziness from moisture in the slip solution often clears within 24-48 hours as the adhesive cures. Don't panic-rework a good install too soon.

Lab 3: Curved Panel Shrink & Install

Lab 3 — Curved Glass Install

Objective: Shrink and install film on a curved practice panel or quarter window.

Time: 90 minutes

1. Place film on exterior curved glass; identify fingers
2. Shrink each finger until film lays flat with no tension
3. Allow to cool; cut pattern with 1/16" margin
4. Transfer to interior; wet-out from center outward
5. Pay attention to edges — light heat may be needed for final tack
6. **Checkpoint:** No fingers remaining; edges fully adhered; no creases

Instructor sign-off: ☐ Pass ☐ Redo

Lab 4: Door Glass Install

Lab 4 — Door Glass Tint Install

Objective: Complete a supervised door glass tint from pattern cut to finished install.

Time: 105 minutes

1. Prep interior glass with full drop or panel access
2. Cut pattern (outside method)
3. Install, wet-out, and trim
4. Inspect under shop lights at multiple angles
5. Document any defects and corrections made

Vehicle assigned: _____

Film used: _____

Instructor sign-off: ☐ Pass ☐ Redo

Day 2 Knowledge Check

1. Why is film shrunk on the outside of the glass before interior installation?
2. Describe the correct heat gun motion during shrinking.
3. What happens if you overheat film during shrinking?
4. How should you squeegee across defroster lines?
5. What is the difference between a "finger" and a bubble?
6. When might you choose to remove a door panel for access?

Day 2 Notes

Day 3 — Professional Practice

Day 3 Learning Objectives

By the end of Day 3, you will be able to:

1. Shrink and install film on a rear window (back glass)
 2. Complete a supervised full-vehicle tint to shop quality standards
 3. Perform a quality control inspection using a standardized checklist
 4. Explain cure time, aftercare, and warranty to a customer
 5. Outline basic pricing structure for tint services
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Day 3 Schedule

Time	Session
8:00 – 8:30	Day 2 review & Q&A
8:30 – 9:30	Classroom: Rear window strategy & windshield brow
9:30 – 10:00	Demo: Rear window shrink overview
10:00 – 10:15	Break
10:15 – 11:00	Classroom: QC inspection & rework decisions
11:00 – 12:00	Lab 5: Rear window install (paired or solo)
12:00 – 1:00	Lunch
1:00 – 1:45	Classroom: Pricing, upselling, and business basics
1:45 – 4:00	Lab 6: Full vehicle tint — supervised capstone install
4:00 – 4:30	Final knowledge check
4:30 – 5:00	Course wrap-up, certificates, feedback

Session 8: Rear Window Strategy

The rear window is the most challenging automotive glass install because of its compound curve. Strategy:

1. **Choose the right film** — some films shrink easier than others; ask your film rep
2. **One-piece vs. seamed** — one-piece looks best; seams are a last resort on extreme curves
3. **Shrink in stages** — tackle the deepest curve areas first
4. **Use plenty of slip** on interior install — rear windows have tight access

5. **H-pattern shrink** — common technique for deep bowl rear windows (instructor demo)

Windshield Brow (Visor Strip)

Where legal, a brow strip on the windshield top is a common add-on:

- Typical depth: 4-6 inches below the shade band
 - Cut from flat film; no shrink usually required
 - Align with rearview mirror sight line — confirm with customer seated in vehicle
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Session 9: Quality Control Inspection

QC Checklist — Every Install

- ☐ Glass prepped — no visible contamination under film
- ☐ Film positioned correctly — no gaps at edges
- ☐ Edges cleanly trimmed — no ragged cuts
- ☐ No fingers on curved areas
- ☐ No bubbles larger than pinhead (customer-visible)
- ☐ Defroster lines intact — none broken or scraped
- ☐ Interior trim and panels reinstalled without damage
- ☐ Customer paperwork completed with film type and VLT documented

Rework Decision Tree

- **Cosmetic defect visible from 3 feet** → rework before delivery
 - **Edge dust speck, customer won't see** → document and deliver with care instructions
 - **Functional issue (edge lift, finger in sight line)** → always rework
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Session 10: Pricing & Business Basics

Common Pricing Models

Model	Description
Per window	Price each position separately
Package	2-front, 4-side, 5-window, 7-window packages
Film tier upsell	Base price for dyed; +\$ for carbon; +\$\$ for ceramic

What to Include in Your Price

- Film cost (waste factor ~10-15%)
- Labor time (your hourly rate × estimated hours)
- Overhead (rent, utilities, insurance, tools)
- Profit margin (target 40-60% gross on labor)

Aftercare Instructions (Give Every Customer)

1. Keep windows up for **48-72 hours** (follow film manufacturer spec)
 2. Clean with ammonia-free cleaner and soft cloth only
 3. Avoid abrasive pads on tinted windows
 4. Small water bubbles are normal during cure — they will dissipate
 5. Contact shop if edge lift occurs within workmanship warranty period
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Lab 5: Rear Window Install

Lab 5 — Rear Window Install

Objective: Shrink and install film on a rear window to shop quality standards.

Time: 60 minutes (instructor may extend as needed)

Vehicle assigned: _____

Shrink technique used: _____

Instructor sign-off: ☐ Pass ☐ Redo

Lab 6: Full Vehicle Capstone Install

Lab 6 — Full Vehicle Tint (Capstone)

Objective: Complete a full-vehicle tint under instructor supervision, meeting all QC checklist items.

Time: 135 minutes

Vehicle assigned: _____

Windows tinted: ☐ Front sides ☐ Rear sides ☐ Rear window ☐ Brow

Film type & VLT: _____

QC checklist completed: ☐ Yes

Instructor final sign-off: ☐ Pass ☐ Redo

Final Knowledge Check

1. Why is the rear window considered the most difficult automotive tint install?
2. List five items on the QC checklist.
3. How long should a customer keep windows up after install?
4. What cleaner should customers avoid on tinted windows?
5. Name three components of a tint pricing structure.
6. When is a rework required vs. acceptable to deliver?

Course Completion Record

Requirement	Completed	Instructor Initials
Day 1 Labs 1-2	☐	_____
Day 2 Labs 3-4	☐	_____
Day 3 Labs 5-6	☐	_____
All knowledge checks	☐	_____
Final capstone install (Pass)	☐	_____

Certificate of Completion issued: ☐ Yes — Date: _____

Instructor signature: _____

Resources & Quick Reference

Slip Solution Quick Mix

1 quart distilled water + 2-4 drops mild soap

Squeegee Direction

Center outward → edges. Defroster lines: parallel strokes.

Cure Time

48-72 hours windows up; full adhesive cure up to 30 days

Film Storage

Cool, dry, upright; avoid direct sunlight on rolls

